Rishab Kolan

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Education

University of California, San Diego — La Jolla, CA
***B.S. in Data Science**; GPA: 3.57* — *Expected June 2028*

Coursework: Principles of Data Science, Programming and Data Structures for Data Science, Data Structures and Algorithms for Data Science, Linear Algebra, Theoretical Foundations of Data Science, Practice of Data Science

Technical Skills

Languages: Java, Python, SQL, R, HTML, CSS, JavaScript, TypeScript
Libraries/Frameworks: React/React Native, Node.js, FastAPI, PyTorch, TensorFlow, scikit-learn
Technologies and Tools: REST API, JUnit, Git, GitHub, PostgreSQL, MySQL, AWS, GCP, Docker

Experience

UC San Diego Health — La Jolla, CA
Data Science Intern — *January 2026 – Present*

- Performed **Random Forest** classical optimization, cross-validation, training, and feature importance analysis on **2000+** bacteria count and metabolic flux features to predict fibrosis stages on **100+** patients with **70%** accuracy
- Performed **PCA** and **PLS-DA** analysis on patient metabolite data to distinguish key metabolites in classifying HIV for **100+** patients with and without liver disease
- Identified **20** key bacteria and metabolites which accelerate fibrosis through spearman correlation heatmaps and feature importance curves, explaining the relationships between bacteria and metabolites in relation to fibrosis

Center for Community Energy — Encinitas, CA
Data Science Intern — *Feb 2026 – May 2026*

- Developed Python scraping and extraction pipelines to collect nonprofit and energy-related data from public sources, IRS filings, and XML records, improving data availability for downstream analysis and reporting
- Built **SQL** datasets by linking organizations through EIN identifiers, merging multi-source nonprofit data, and standardizing records to create reliable, analytics-ready data for federal solar incentive outreach initiatives
- Improved data coverage and quality by integrating external data sources, cleaning and validating contact information, and automating data preparation workflows to support targeting, analysis, and decision-making

Triton NeuroTech @ UC San Diego — La Jolla, CA
Project Member — *November 2025 – April 2026*

- Built an EMG data collection and preprocessing pipeline using MindRove Connect, LibEMG, and SciPy for signal filtering, normalization, and sequence generation from multi-channel biosignal data
- Designed an **LSTM** based gesture-recognition model built in **TensorFlow** and evaluating GWO for optimization
- Outlined a real-time gesture-to-movement mapping system with thresholding to reduce false triggers

Projects

Fragrance Recommender | *Website* — *October 2026*

- Built and deployed a full-stack fragrance recommendation platform using Python, **FastAPI**, **Next.js**, **TypeScript**, and **PostgreSQL**, processing **140,000+** fragrances
- Developed a personalized recommendation engine using **TF-IDF**, **cosine similarity**, clustering, and **Bayesian ratings**, generating 10 recommendation categories with custom filters and clone exclusions
- Deployed containerized services on **Google Cloud** Run with Neon PostgreSQL, **Google OAuth**, Secret Manager, and Cloud Monitoring, supporting saved collections and cached Amazon pricing

Posterboy | *Website* — *July 2026*

- Developed a full-stack **React/TypeScript** web app that analyzes listening history through Spotify API and recommends personalized album posters through a **FastAPI/Python** backend and **PostgreSQL** database
- Engineered a Spotify **OAuth 2.0** workflow and reciprocal-rank scoring pipeline to aggregate users' top albums
- Built a **scikit-learn** K-Means pipeline to recommend album covers with visually similar color palettes
- Containerized the full-stack application with **Docker** and deployment through GitHub Actions, **Vercel**, and **GCP**